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ReAttnCLIP: Training-Free Open-Vocabulary Remote Sensing Image Segmentation via Re-defined Attention in CLIP

Xin Niu[1,2,3*†] Manqi Zhao[5*] Dongsheng Jiang[5*] Yingying Wu[4] Bing Su[1,2,3‡]
[1]Gaoling School of Artificial Intelligence, Renmin University of China
[2]Beijing Key Laboratory of Research on Large Models and Intelligent Governance
[3]Engineering Research Center of Next-Generation Intelligent Search and Recommendation, MOE
[4]BeiJing China-Power Information Technology Co.,Ltd [5]Huawei Technologies Co., Ltd

niuxin@ruc.edu.cn, 933475@gmail.com, dongsheng_jiang@outlook.com
smile.cherokee@gmail.com, subingats@gmail.com

Abstract

Remote sensing image segmentation is essential for applications such as natural disaster monitoring and precision agriculture. Open-vocabulary segmentation improves flexibility by removing fixed category constraints, enabling more fine-grained scene understanding. However, unlike CLIP's pretraining objective that emphasizes global image–text alignment, segmentation requires discriminative patch-level representations for accurate pixel-wise prediction. Consequently, the quality of attention maps in the final transformer layers is critical for modeling interactions among spatial regions. Existing methods often produce suboptimal representations when capturing the complex spatial structures of remote sensing imagery. To address this issue, we refine CLIP's attention mechanism through three modifications: (1) replacing patch-to-patch attention with intermediate-layer feature similarities to better preserve spatial structure; (2) leveraging intermediate-layer attention for class-to-patch alignment to reduce classification interference; and (3) disabling the `[CLS]` token's self-attention to mitigate bias. Experiments on multiple remote sensing benchmarks, including building and road extraction datasets, show that our method achieves state-of-the-art performance among training-free approaches.

1. Introduction

Remote sensing image segmentation plays a crucial role in various fields, such as environmental monitoring, disaster assessment, urban planning, and more [12]. Traditional remote sensing image segmentation methods typically rely on extensive pixel-level annotations for training [39], which can address some segmentation challenges to a certain extent. Although effective in controlled scenarios, these methods show limited adaptability to the inherent complexities of remote sensing images, such as heterogeneous landscapes, extreme scale variations, and complex target distributions, leading to compromised generalization capabilities [16]. Furthermore, these methods struggle to meet the dynamic real-world demands, as frequently emerging unannotated categories make the traditional closed-vocabulary

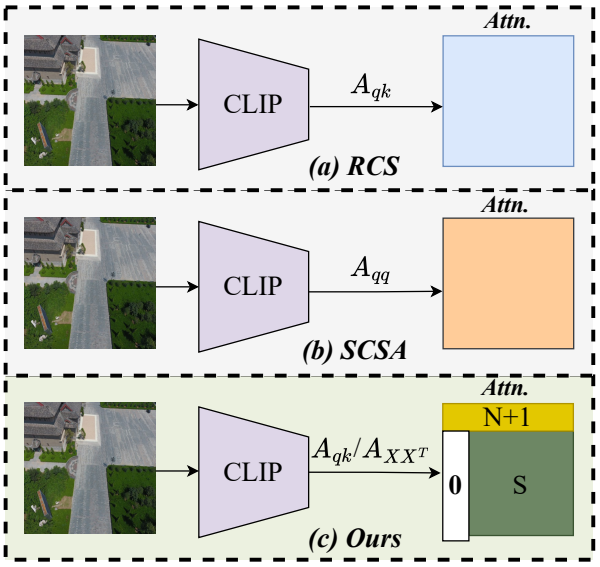


Figure 1. Comparison between the proposed method and existing approaches. (a) Residual Cross Attention (RCS); (b) Self-Cross Self-Attention (SCSA); (c) the proposed Refine Attention. Wherein ***Attn.*** denotes the attention map.

*These authors contributed equally.
†Work done during an internship at Huawei.
‡Corresponding author.

paradigm inadequate.

To mitigate these constraints, Open-Vocabulary Segmentation has emerged, integrating semantic reasoning to segment arbitrary objects via natural language descriptions without category-specific retraining . This paradigm reduces dependency on exhaustive annotated datasets and accommodates unseen categories. Nevertheless, most existing open-vocabulary methods [8, 26, 38, 41] still require fine-tuning with curated data, introducing deployment bottlenecks and performance inconsistencies in practical remote sensing applications, especially in scenarios where annotated resources are scarce.

With the advancement of training-free segmentation methods [1, 23, 34], they bypass the optimization of specific datasets by leveraging large-scale pre-trained models, offering a paradigm shift. These methods directly extract transferable features from the base models, significantly reducing or eliminating training overhead. Despite the progress made in certain fields, the dependence of most models on the training process restricts the application of training-free methods in real-world scenarios. To address this shortcoming, numerous improvements based on the CLIP [28] (Contrastive Language-Image Pretraining) model have gained widespread attention in recent years.

CLIP's image-text pre-training excels in overall image understanding, but its operations are at the image level, which fundamentally misaligns with the pixel-level discriminative features required for segmentation. This discrepancy is particularly prominent in the remote sensing field, where complex scene distributions, extreme scale variations, and heterogeneous landscapes demand fine-grained feature representations. Current adaptation methods typically use attention mechanisms to derive dense features, generating token embeddings by re-calculating the query-key (RCS) or key-key (SCSA) attention map [19]. By treating attention maps as monolithic entities and focusing on restructuring rather than component-wise analysis, these methods fail to balance global semantics and local details, leading to suboptimal representations for remote sensing imagery.

In this paper, we innovatively conduct a comprehensive analysis of CLIP's attention mechanisms for remote sensing segmentation, decomposing their influence into three interconnected dimensions (As show in Fig. 1): (1) the similarity calculation between patch tokens, modeling inter-region relationships; (2) the composition of the `[CLS]` token, capturing global information; and (3) the `[CLS]` token's contribution to patch tokens, revealing its impact on local regions during segmentation. Our method strategically recalibrates these elements, enabling precise open-vocabulary segmentation without training, while directly addressing the inherent complexity of remote sensing.

In summary, our contributions are as follows:

- We provide the first detailed analysis of how CLIP's attention map components impact remote sensing image segmentation, focusing on three key elements: patch token similarity, the `[CLS]` token composition, and its contribution to patch tokens.
- We propose a novel approach to effectively extract high-quality dense features from CLIP's attention map for pixel-level image segmentation, addressing the challenge of transforming CLIP's image-level features into dense, pixel-wise representations.
- We conducted experiments on ten public remote sensing datasets, achieving state-of-the-art performance with an average gain of 1.7% on open-vocabulary segmentation and 1.1% on object extraction tasks.

2. Related Work

2.1. Vision-Language Model

Vision-language models [6, 10, 15, 27] establish connections between visual and textual modalities, and these models are typically pretrained on large-scale image-text datasets [29]. Among these models, CLIP is one of the most widely adopted. CLIP is a model trained using contrastive learning on a large collection of image-text pairs, enabling it to perform zero-shot retrieval tasks effectively. Specifically, the classification result is obtained by computing the similarity between the output `[CLS]` token of the vision encoder and the embedding produced by the text encoder. However, due to the limitations of its pretraining strategy, the model can only perform image-level alignment with text. It cannot be directly applied to pixel-level alignment tasks, such as segmentation or object detection. Many existing methods [31, 36] have attempted to modify the CLIP model to enhance its fine-grained feature extraction capabilities while preserving its strong generalization ability.

2.2. Open-Vocabulary Natural Image Segmentation

Open-vocabulary segmentation aims to detect and label image regions based on a list of vocabulary terms. Applying CLIP to dense prediction tasks presents challenges. MaskCLIP [11, 43] highlights that CLIP's image features are primarily determined by the value (V) component in the attention mechanism, proposing to replace the query (Q) and key (K) matrices with a 1×1 convolution. While CLIP's direct output for segmentation often leads to suboptimal performance, it still provides useful guidance. CLIP Surgery [21] suggests refining these coarse predictions using prompts like points or boxes, which are then input into segmentation models like SAM [17].

SCLIP [32] analyzes CLIP's final attention block and shows that query-query similarity improves dense visual representations by capturing relationships between patch tokens of the same category. NACLIP [13] refines this with

a Gaussian kernel to enhance similarity measurements between patch tokens. ClearCLIP [18] removes residual connections in CLIP's final block to reduce noise, improving patch embeddings. In contrast, CLIPTrase [30] explores intermediate layer outputs, using clustering to generate high-quality masks for guiding segmentation. ResCLIP [40] demonstrates that intermediate layer attention maps can significantly enhance patch token representations.

2.3. Open-Vocabulary Remote Sensing Image Segmentation

Although numerous methods have adapted CLIP for segmentation, most are developed for natural images, as CLIP is pretrained on large-scale natural image–text pairs. When applied to remote sensing imagery, these approaches often suffer from severe domain discrepancies. To address this issue, several training-based open-vocabulary remote sensing segmentation methods, such as OVRS [5] and GSNet [41], have been proposed to improve performance through task-specific learning. In contrast, SegEarth-OV [20] attempts to bridge the gap between CLIP pretraining and downstream segmentation by introducing a learnable upsampling module and a heuristic strategy to reduce the influence of the `[CLS]` token on patch tokens. However, existing methods generally treat the attention map as a whole without analyzing its internal components. In this work, we conduct an in-depth analysis of the attention computation process and introduce targeted refinements for each component of the attention map.

3. Preliminaries

3.1. Dense Visual Feature Extraction with CLIP

CLIP is a vision–language model trained on large-scale image–text pairs, consisting of an image encoder and a text encoder. The image encoder adopts a ViT-B/16 architecture that partitions an image $I \in \mathbb{R}^{H\times W\times 3}$ into $N = \frac{H}{16} \times \frac{W}{16}$ patches. A classification token is prepended to form

$$X = [x_{\text{cls}}, x_1, \ldots, x_N] \in \mathbb{R}^{(N+1)\times d}. \tag{1}$$

Self-attention is computed as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V, \tag{2}$$

where $Q = XW^Q$, $K = XW^K$, and $V = XW^V$, producing the attention map

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right). \tag{3}$$

While CLIP uses x_{cls} for global representation, dense prediction relies on patch tokens $x_1, \ldots, x_N$. Therefore, analyzing the attention map is crucial for effectively leveraging these spatial tokens.

3.2. Alleviating the Impact of Global Representations

All patch tokens in CLIP interact with the `[CLS]` token via self-attention, introducing global information that can reduce their usefulness for dense prediction. SegEarth-OV mitigates this by subtracting the `[CLS]` embedding from each patch:

$$X = [x_{\texttt{cls}}, x_1, \ldots, x_N] \in \mathbb{R}^{(N+1)\times d}, \tag{4}$$

$$\tilde{x}_i = x_i - x_{\texttt{cls}}, \quad i = 1, \ldots, N. \tag{5}$$

The resulting patch features $\tilde{X} = [\tilde{x}_1, \ldots, \tilde{x}_N]$ are less influenced by global context, improving local discriminability.

4. Method

In this section, we decompose the attention map into three distinct components and analyze the influence of each on the patch token representations. For each component, we further propose targeted optimization strategies to enhance the quality of the resulting dense features. Following the setup of SegEarth-OV [20], we also use SimFeatUp as the upsampling module. The overview is shown in Fig. 2.

4.1. Motivation for Attention Map Analysis

Consider the standard attention update in the final CLIP transformer block:

$$X = AV, \quad V = [v_{\text{CLS}}, v_1, v_2, \ldots, v_{196}], \tag{6}$$

where $A \in \mathbb{R}^{197\times 197}$ denotes the attention map and V the value matrix. Expanding row-wise, the output of the i-th patch token can be written as

$$x_i = A_{i0} v_{\text{CLS}} + \sum_{j=1}^{196} A_{ij} v_j. \tag{7}$$

From Eq. (7), each patch embedding x_i receives information from two sources: the global [CLS] token (via A_{i0}) and other patch tokens (via the patch–patch submatrix of A).

Prior studies[20] have shown that the [CLS] token can introduce global bias in dense prediction tasks, degrading patch-level representations. To mitigate this, we adopt two complementary strategies: (1) zeroing the first column of the attention map to block direct [CLS] influence, and (2) subtracting a reconstructed [CLS] component to eliminate residual effects.

Based on these observations, we decompose the attention map into three interpretable parts: **(i)** [CLS]→patch attention, **(ii)** patch→[CLS] attention, and **(iii)** patch↔patch attention. This decomposition enables explicit control and analysis of distinct information flows in CLIP's attention, leading to more localized and precise dense predictions.

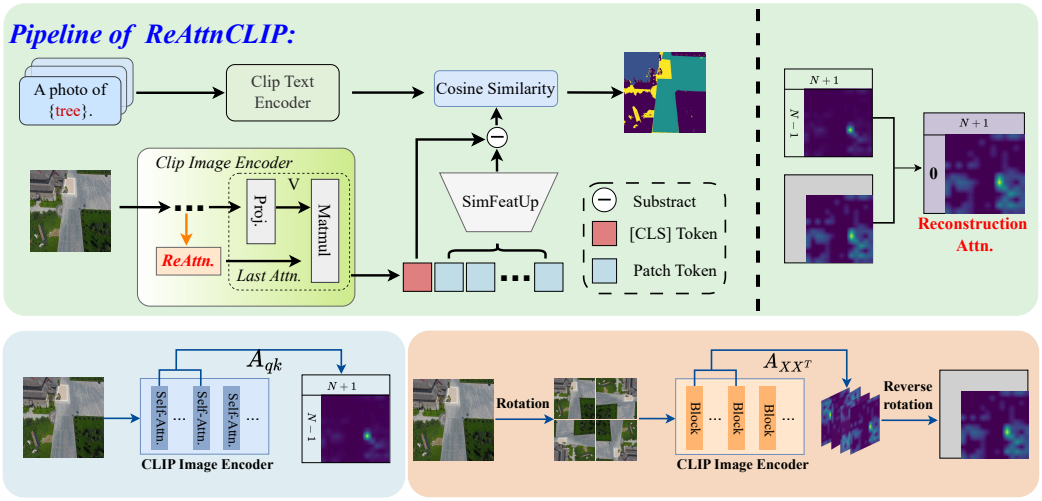


Figure 2. Overview of ReAttnCLIP. ReAttnCLIP enhances dense feature representation by strategically modifying the attention map in CLIP's final block, decomposing it into three components. The second row illustrates the construction of Patch–Patch relationships and the formation of the [CLS] token. The right half of the first row demonstrates how the [CLS] token's contribution to patch tokens is set to zero. Additionally, SimFeatUp, a trained upsampling module specifically designed for remote sensing images, further improves feature resolution and segmentation performance.

4.2. Patch-to-Patch Similarity

For a standard ViT, the attention map is 197 × 197, and removing the first row and column for the [CLS] token yields a 196 × 196 matrix of patch-patch similarities. Recent works have proposed strategies to improve these maps: SegEarth-OV combines Query-Query, Key-Key, and Value-Value similarities via a weighted sum for enhanced attention:

$$A = \alpha \cdot (QQ^\top) + \beta \cdot (KK^\top) + \gamma \cdot (VV^\top), \quad (8)$$

where Q, K, V denote the query, key, and value matrices, and α, β, γ are weighting coefficients.

The advantage of Query-Query similarity lies in its symmetric structure, which avoids discrepancies introduced by separate projections in standard Query-Key attention. This reduces projection-induced bias and enhances the reliability of token relationships, which is particularly beneficial for dense prediction tasks such as segmentation.

Motivated by these insights, we further simplify the attention computation by removing all learnable projections and directly measuring similarity between raw patch embeddings:

$$\text{Attention}_{i,j}^{\text{raw}} = \frac{x_i^\top x_j}{\sqrt{d}}, \quad (9)$$

where x_i, x_j are the original patch embeddings. This projection-free approach provides an interpretable baseline to assess the intrinsic geometric relationships in the embedding space for segmentation.

We further explore features from intermediate CLIP layers and visualize attention maps generated by different strategies, as illustrated in Fig. 3. The results show that computing attention directly from patch token embeddings achieves competitive performance.

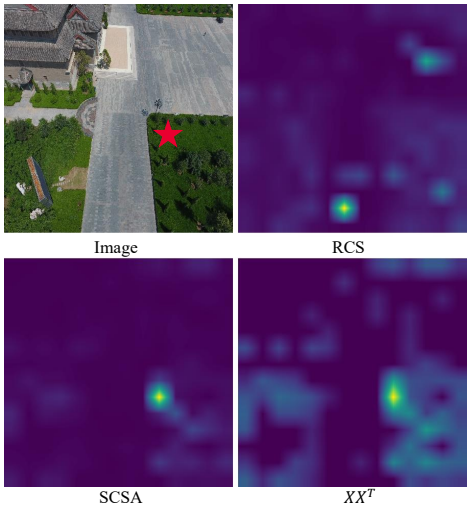


Figure 3. From left to right and top to bottom, the figure visualizes RCS, SCSA and $XX^\top$.

Remote sensing images often present significant variations in object orientation, making robust feature extraction challenging. To address this, we apply rotational augmentation by rotating each image at multiple angles (e.g., $0°, 90°, 180°, 270°$), extracting patch embeddings

from each rotation, and computing the corresponding similarity matrices for each rotation. Specifically, for each selected layer $k \in \mathcal{K}$, we compute the patch embeddings similarity as follows:

$$S^{(r,k)} = X^{(r,k)} X^{(r,k)\top}, \tag{10}$$

where $X^{(r,k)} \in \mathbb{R}^{N \times d}$ represents the patch embeddings for the r-th rotation at layer k.

The final rotation-enhanced similarity is obtained by weighted aggregation across all rotations and averaging over layers:

$$S_{\text{rot}} = \frac{1}{|\mathcal{K}|} \sum_{k \in \mathcal{K}} \sum_{r} \lambda_r \cdot S^{(r,k)}, \tag{11}$$

where λ_r denotes the weight for each rotation, and $\mathcal{K}$ is the set of selected layers. The averaging over layers k ensures that the rotation-enhanced similarity captures information across multiple layers.

The overall similarity matrix is then computed by combining the rotation-enhanced similarity with attention maps from selected layers:

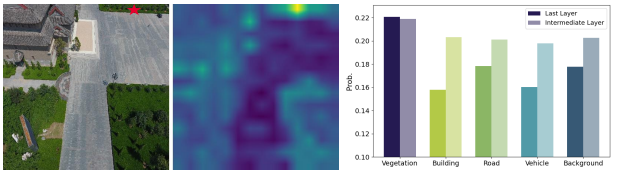
$$S = \alpha \cdot S_{\text{rot}} + \sum_{l \in \mathcal{L}} \beta_l \cdot A^{(l)}, \tag{12}$$

where $A^{(l)}$ denotes the attention map from layer l (which corresponds to the Query-Key attention map), and $\mathcal{L}$ is the set of selected layers. S_{rot} represents the similarity derived from XX^T attention maps, which captures the pairwise similarities between patch embeddings.

4.3. CLS Token Reconstruction

When adapting CLIP to dense prediction tasks, the `[CLS]` token is typically discarded, and only patch tokens are used. However, during pretraining, patch tokens interact with the `[CLS]` token, leading to patch tokens encoding significant global context. To mitigate this, SegEarth-OV subtracts the final-layer `[CLS]` token embedding from each patch token to remove global bias. In fact, patch tokens also carry information from other categories during this attention computation process. The `[CLS]` token in the final layer carries only a small amount of category information and does not encompass information from all categories.

As shown in Fig. 4, we visualize the similarity between patch tokens based on their representations derived from the value projection in the final-layer self-attention. It can be observed that the selected patch token corresponds to the "vegetation" category; however, the region associated with "building" is also activated. This indicates that the patch token carries information from multiple categories. Furthermore, we visualize the similarity between the output `[CLS]` token and the text embeddings. The results show



(a) Similarity of patch tokens (b) Similarity of output [CLS] token and text

Figure 4. Visualization of the similarities between patch tokens from the final transformer layer and the similarities between text embeddings and `[CLS]` tokens derived from different layers.

that the `[CLS]` token constructed using attention maps from intermediate layers contains more diverse category information.

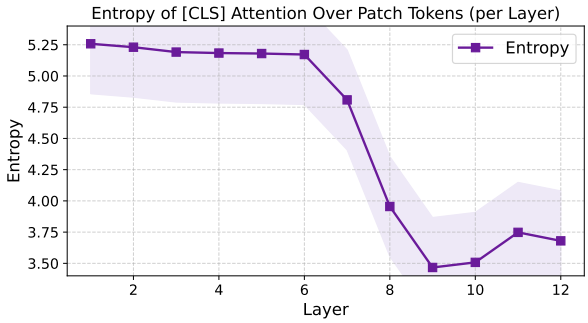


Figure 5. We visualized the entropy of the attention maps of the `[CLS]` token across different layers, which represents the extent of interaction between the `[CLS]` token and the other patch tokens.

As shown in Fig. 5, we visualize the entropy of the attention maps to examine the interactions between the `[CLS]` token and the other patch tokens across different layers. It can be observed that as the layer depth increases, the entropy gradually decreases, indicating that the attention of the `[CLS]` token shifts towards a subset of patch tokens. Based on our observations, the `[CLS]` token embedding at this stage may contain only a limited amount of information.

Inspired by this, we propose reconstructing the global representation by aggregating the attention distributions from intermediate layers. Specifically, for each selected intermediate layer $l \in \mathcal{L}$, we extract the entire first row of the attention map $A^{(l)} \in \mathbb{R}^{(N+1)\times(N+1)}$, corresponding to the attention weights from the `[CLS]` token to all tokens, including itself. The average of these rows across the selected layers forms a composite attention vector:

$$\bar{A}^{(l)}_{0,:} = \frac{1}{|\mathcal{L}|} \sum_{l \in \mathcal{L}} A^{(l)}_{0,:}, \tag{13}$$

where $\bar{A}^{(l)}_{0,:} \in \mathbb{R}^{N+1}$ denotes the mean attention weights

Table 1. Comparison Results on Open-Vocabulary Remote Sensing Segmentation Datasets (mIoU).

Methods	OpenEarthMap	LoveDA	iSAID	Potsdam	Vaihingen	UAVidimg	UDD5	VDD	Average
CLIP *(ICML21)*	12.0	12.4	7.5	14.5	10.3	10.9	9.5	14.2	11.4
MaskCLIP *(ECCV22)*	25.1	27.8	14.5	31.7	24.7	28.6	32.4	32.9	27.2
SCLIP *(ECCV24)*	29.3	30.4	16.1	36.6	28.4	31.4	38.7	37.9	31.1
GEM *(CVPR24)*	33.9	31.6	17.7	36.5	24.7	33.4	41.2	39.5	32.3
ClearCLIP *(ECCV24)*	31.0	32.4	18.2	40.9	27.3	36.2	41.8	39.3	33.4
ResCLIP *(CVPR25)*	34.3	29.6	8.8	42.6	27.8	36.0	41.9	39.6	32.6
SegEarth-OV *(CVPR25)*	40.3	36.9	21.7	47.1	29.1	42.5	50.6	45.3	39.2
Ours	**41.1**	**37.0**	**23.2**	**48.7**	**29.9**	**44.0**	**53.7**	**49.7**	**40.9**

from the [CLS] token to all tokens (including itself), averaged over the selected intermediate layers. This vector is then used as a refined global attention distribution for downstream debiasing operations.

This strategy yields a global representation that integrates both spatial and semantic information across the network's depth, providing a more comprehensive and robust debiasing reference for downstream tasks.

4.4. CLS-to-Patch Attention Influence

The [CLS] token's contribution to each patch is given by the first column of the attention matrix, reflecting its influence on patch representations. Such global-to-local interactions can introduce undesired bias at the patch level.

As noted above, the [CLS] token already biases patch embeddings during pretraining. We therefore set this contribution to zero to further reduce its effect on local features.

$$A_{refine}[i, 0] = 0, \quad \text{for } i = 1, 2, \ldots, N, \tag{14}$$

where $A_{refine} \in \mathbb{R}^{(N+1)\times(N+1)}$ is the redefined attention matrix, rather than the attention map, and N is the number of patch tokens.

To summarize, we redefine the structure of the attention map in the last block of CLIP by explicitly decomposing it into three distinct components corresponding to different spatial regions within the attention matrix:

$$A_{refine} = \begin{bmatrix} \bar{A}^{(l)}_{0,0} & \bar{A}^{(l)}_{0,1:196} \\ 0 & S \end{bmatrix}, \tag{15}$$

where $S \in \mathbb{R}^{N\times N}$ denotes the patch-to-patch similarity matrix (i.e., the lower-right submatrix), $\bar{A}^{(l)}_{0,:} \in \mathbb{R}^{1\times N+1}$ represents the attention from the [CLS] token to all patch tokens (i.e., the first row excluding the top-left element).

5. Experiment

5.1. Datasets and Metrics

To validate the effectiveness of the proposed method, we conduct evaluations on three tasks: open-vocabulary semantic segmentation of remote sensing images, building extraction, and road extraction. Specifically, we evaluate our method on eight publicly available datasets for the open-vocabulary segmentation task, and on one dataset each for the building and road extraction tasks. Overall, our method is assessed on ten datasets across three distinct tasks.

Datasets. For the open-vocabulary segmentation task, we select eight publicly available datasets, namely OpenEarthMap [37], LoveDA [33], iSAID [35], Potsdam, Vaihingen, UAVid [24], UDD5 [7], and VDD [4]. For building extraction and road extraction tasks, we selected one dataset for each, namely WHU$^{SAT.II}$ [14] and Massachusetts [25].

Metrics. We follow prior work [22] by adopting the Intersection-over-Union (IoU) as the evaluation metric and apply it consistently across all models.

5.2. Implementation Details

We adopt ViT/B as the backbone model. All input images are resized to 448×448 before being fed into the CLIP image encoder. During inference, we apply a sliding window strategy to preserve model performance. Specifically, we use a window size of 224×224 to slide across the image and then stitch the predicted patches together to form the final segmentation result of the original image. For the text input, we follow OpenAI's approach by averaging the embeddings generated from 80 different templates. For the parameter l, we define it to be between layers 6 and 9, its value is determined for each dataset based on the entropy map of the [CLS] token. λ_0 is set to 1, and λ_i for $i \in \{1, 2, 3\}$ is set to 0.4. $\mathcal{K}$ contains layers 9 to 11. The parameter α is set to 0.2. All experiments were conducted on a single NVIDIA V100 GPU.

5.3. Compared with state-of-the-art methods

We compare our approach against six state-of-the-art training-free open-vocabulary segmentation methods across three tasks: CLIP [28], MaskCLIP [43], SCLIP [32], GEM [2], ClearCLIP [18], ResCLIP[40] and SegEarth-OV [20].

Semantic segmentation. The results on semantic segmentation datasets are presented in Table 1. Our method achieves the new state-of-the-art performance across all eight datasets. Compared with the current state-of-the-art

Table 2. Comparison Results on Remote Sensing Building/Road Extraction Datasets (mIoU).

Methods	WHU$^{SAT.II}$	Massachusetts	Average
CLIP	3.5	4.9	4.2
MaskCLIP	14.0	10.6	12.3
SCLIP	21.0	7.4	14.2
GEM	13.6	5.1	9.4
ClearCLIP	20.8	6.4	13.6
ResCLIP	14.8	4.9	9.9
SegEarth-OV	28.4	11.5	20.0
Ours	**29.7**	**12.4**	**21.1**

method SegEarth-OV. Our method achieves an improvement of 3.1% and 4.4% in mIoU on the UDD5 and VDD datasets, respectively. On the remaining six datasets, our method also achieves improvements of approximately 1% or more. The performance improvement on the LoveDA dataset is limited to 0.1%. This is mainly because the images in this dataset are generally blurry and contain mostly small objects. Since we use a ViT backbone pre-trained on natural images, there exists a noticeable domain gap between natural and remote sensing imagery, which restricts the effectiveness of our method on this particular dataset.

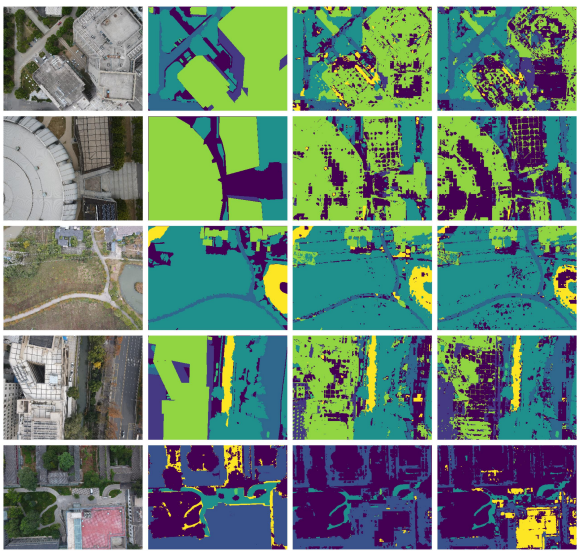


Figure 6. Qualitative Comparison of Segmentation Results. The visualization results are presented from left to right as the original image, ground truth, our method, and the current state-of-the-art method.

Building/Road extraction. To evaluate the generalization ability of the proposed method, we further validate it on two distinct object extraction datasets. In the remote sensing domain, road extraction is generally more challenging due to the irregular and highly variable shapes of roads, particularly for training-free approaches. Nevertheless, our method achieves approximately 1% improvement over the current state-of-the-art on the road extraction dataset.

Qualitative Results. The qualitative results are shown in Fig. 6. It can be observed that our method yields segmentation maps that are more consistent with the ground truth. Specifically, our approach better preserves object boundaries and captures fine-grained details, such as narrow roads and small structures, which are often over-smoothed or misclassified by SegEarth-OV.

5.4. Ablation Study and Analysis

Ablation Study of Different Modules. We conducted ablation studies to evaluate the effectiveness of the proposed method, with results shown in Table 3. Enabling the P-P (Patch-to-Patch Similarity) module alone consistently improves performance across all three datasets, emphasizing the importance of patch-level similarity modeling. Incorporating the CLS token adjustment further enhances performance by reconstructing a more informative global representation. The CLS-Patch module, which reduces the global token's residual influence on patch tokens, also provides additional benefits when applied independently.

Table 3. Exploring the effectiveness of the proposed different components. We conducted experiments to validate the results of different components. P-P represents the similarity between patch tokens, CLS refers to the construction of the `[CLS]` token, and CLS-Patch denotes the contribution of the `[CLS]` token to the patch tokens.

P-P	CLS	CLS-Patch	UDD5	VDD	WHU$^{SAT.II}$
			50.4	45.3	28.4
✓			53.1	47.8	29.0
	✓		52.5	46.8	28.9
		✓	51.5	46.7	28.8
✓	✓	✓	**53.7**	**49.7**	**29.5**

Ablation Study on the P-P Module. The experimental results demonstrate that the P-P module is crucial for performance. We further investigate the impact of different computation strategies within the P-P module, as shown in Table 4. Computing similarity using $XX^\top$ yields the most significant improvement, with a 2.8% increase on the UDD5 dataset. Additionally, rotation operations consistently enhance performance across all datasets, with the largest gain observed on the VDD dataset, where small objects benefit from rotation. To further enrich the attention map, we incorporate additional attention maps from intermediate Query-Key computations, leading to further performance gains.

Ablation study on different layers. As shown in Table 5, we investigate the impact of different layer selections for the $\mathbf{XX}^\top$ Fusion strategy. The results indicate that varying

Table 4. Ablation study on the P-P module. We investigate the impact of different computation strategies.

$\mathbf{XX}^\top$	$\mathbf{QK}^\top$	Rotation	UDD5	VDD	$\text{WHU}^{SAT.II}$
			50.4	45.3	28.4
✓			53.2	46.7	29.1
	✓		52.7	47.5	28.7
✓		✓	53.4	48.8	29.3
✓	✓	✓	**53.7**	**49.7**	**29.5**

the input layers leads to only marginal differences in performance on both the UDD5 and VDD datasets, demonstrating the robustness of the proposed fusion mechanism. To balance accuracy and computational efficiency, we adopt the fusion of layers 9 to 11 as our default setting.

Table 5. Analysis of layer fusion strategies.

Fusion Ways	Layers	UDD5	VDD
$\mathbf{XX}^\top$ *Fusion*	$7 \rightarrow 9$	53.7	49.6
	$7 \rightarrow 10$	53.7	49.6
	$7 \rightarrow 11$	53.7	49.7
	$8 \rightarrow 11$	**53.8**	49.6
	$9 \rightarrow 11$	53.7	49.7
	$10 \rightarrow 11$	53.4	**49.8**

Complexity Analysis. As shown in Table 6, our method achieves the highest overall segmentation performance (40.9 Average) while maintaining a reasonable balance between inference latency and computational cost. Although the latency and FLOPs are moderately higher than those of previous methods, the substantial improvement in accuracy demonstrates that our approach effectively enhances model capability without excessive computational overhead.

Table 6. Comparison of inference latency and computational cost (GFLOPs) across different methods.

Method	Inference Latency	GFLOPS	Average
MaskCLIP	9.5 ms	12.2	27.2
ClearCLIP	20.0 ms	11.7	33.4
ResCLIP	70.2 ms	10.4	32.6
SegEarth-OV	22.8 ms	27.8	39.2
Ours	52.3 ms	60.2	40.9

Plug and Play. Table 7 shows that integrating our plug-and-play module into existing open-vocabulary segmentation methods consistently boosts performance across multiple remote sensing datasets. Notably, when applied to CLIP-based frameworks such as MaskCLIP, SCLIP, and ClearCLIP, the module yields substantial improvements, with increases of up to 9.5% in the average score, demonstrating its general effectiveness across different baseline models.

Table 7. Effect of the proposed plug-and-play module on open-vocabulary remote sensing segmentation.

Methods	UDD5	VDD	Potsdam	Average
MaskCLIP	32.4	32.9	31.7	32.3
+ Ours	42.5	40.4	42.5	41.8 ↑9.5
SCLIP	38.7	37.9	36.6	37.7
+ Ours	43.0	40.7	42.9	42.2 ↑4.5
ClearCLIP	39.3	31.0	40.9	37.1
+ Ours	48.4	45.5	44.3	46.1 ↑9.0

Results on natural image datasets. Table 8 presents a comparison of open-vocabulary segmentation methods on three natural image datasets: Stuff[3], ADE20K[42], and Cityscapes[9]. For each baseline method—MaskCLIP, SCLIP, and ClearCLIP—the table shows performance both before and after integrating our plug-and-play module. The improvements achieved by our module are highlighted in green in the Average column, illustrating consistent gains across all datasets and demonstrating its effect on enhancing segmentation accuracy for each method.

Table 8. Performance comparison of open-vocabulary segmentation methods on natural images.

Methods	Stuff	ADE20K	Cityscapes	Average
MaskCLIP	16.4	9.8	12.6	12.9
+ Ours	22.1	16.6	29.0	22.6 ↑9.7
SCLIP	21.1	14.6	29.1	21.6
+ Ours	23.4	16.8	30.8	23.7 ↑2.1
ClearCLIP	23.9	16.7	30.0	23.5
+ Ours	24.2	17.3	31.2	24.2 ↑0.7

6. Conclusion

We introduce a novel framework for open-vocabulary segmentation of remote sensing images by utilizing attention maps derived from CLIP. A comprehensive analysis is conducted to examine the impact of various components within CLIP's attention mechanism on segmentation accuracy, along with a methodology to extract high-quality dense features for pixel-level segmentation. Specifically, we investigate the evolution of attention patterns across network layers and their role in enhancing spatial localization. Extensive experiments on multiple benchmark datasets demonstrate that our method achieves state-of-the-art performance and exhibits strong generalization across diverse data sources. The effectiveness and robustness of attention-based strategies for efficient remote sensing image segmentation are thereby validated.

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